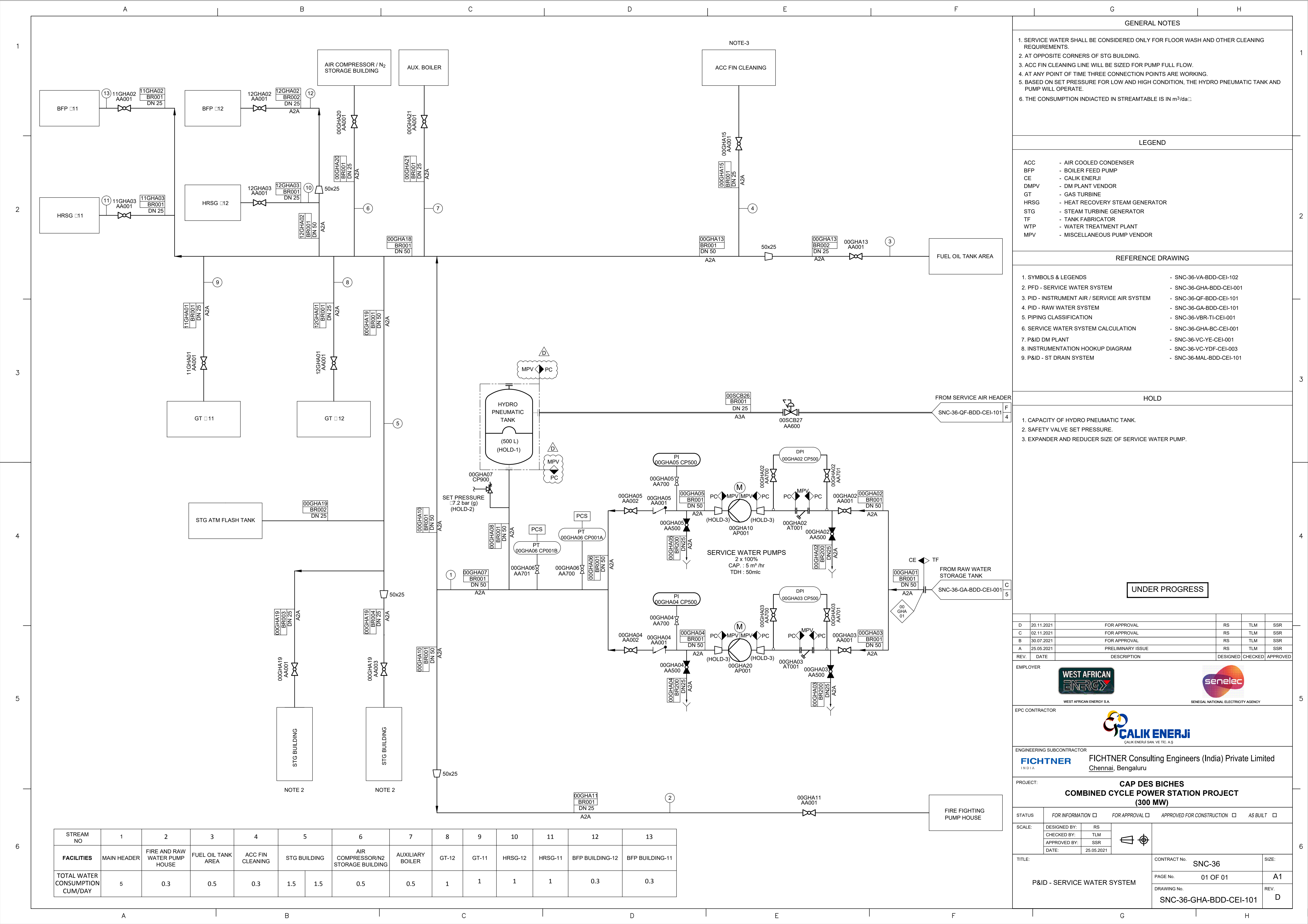




- GENERAL NOTES
- SERVICE WATER SHALL BE CONSIDERED ONLY FOR FLOOR WASH AND OTHER CLEANING REQUIREMENTS.
 - AT OPPOSITE CORNERS OF STG BUILDING.
 - ACC FIN CLEANING LINE WILL BE SIZED FOR PUMP FULL FLOW.
 - AT ANY POINT OF TIME THREE CONNECTION POINTS ARE WORKING.
 - BASED ON SET PRESSURE FOR LOW AND HIGH CONDITION, THE HYDRO PNEUMATIC TANK AND PUMP WILL OPERATE.
 - THE CONSUMPTION INDICATED IN STREAMTABLE IS IN m³/da.

- LEGEND
- ACC - AIR COOLED CONDENSER
 - BFP - BOILER FEED PUMP
 - CE - CALIK ENERJİ
 - DMPV - DM PLANT VENDOR
 - GT - GAS TURBINE
 - HRSG - HEAT RECOVERY STEAM GENERATOR
 - STG - STEAM TURBINE GENERATOR
 - TF - TANK FABRICATOR
 - WTP - WATER TREATMENT PLANT
 - MPV - MISCELLANEOUS PUMP VENDOR

- REFERENCE DRAWING
- | | |
|--|--------------------------|
| 1. SYMBOLS & LEGENDS | - SNC-36-VA-BDD-CEI-102 |
| 2. PFD - SERVICE WATER SYSTEM | - SNC-36-GHA-BDD-CEI-001 |
| 3. PID - INSTRUMENT AIR / SERVICE AIR SYSTEM | - SNC-36-QF-BDD-CEI-101 |
| 4. PID - RAW WATER SYSTEM | - SNC-36-GA-BDD-CEI-101 |
| 5. PIPING CLASSIFICATION | - SNC-36-VBR-TI-CEI-001 |
| 6. SERVICE WATER SYSTEM CALCULATION | - SNC-36-GHA-BC-CEI-001 |
| 7. P&ID DM PLANT | - SNC-36-VC-YE-CEI-001 |
| 8. INSTRUMENTATION HOOKUP DIAGRAM | - SNC-36-VC-YDF-CEI-003 |
| 9. P&ID - ST DRAIN SYSTEM | - SNC-36-MAL-BDD-CEI-101 |

- HOLD
- CAPACITY OF HYDRO PNEUMATIC TANK.
 - SAFETY VALVE SET PRESSURE.
 - EXPANDER AND REDUCER SIZE OF SERVICE WATER PUMP.

UNDER PROGRESS

REV.	DATE	DESCRIPTION	DESIGNED	CHECKED	APPROVED
D	20.11.2021	FOR APPROVAL	RS	TLM	SSR
C	02.11.2021	FOR APPROVAL	RS	TLM	SSR
B	30.07.2021	FOR APPROVAL	RS	TLM	SSR
A	25.05.2021	PRELIMINARY ISSUE	RS	TLM	SSR

EMPLOYER



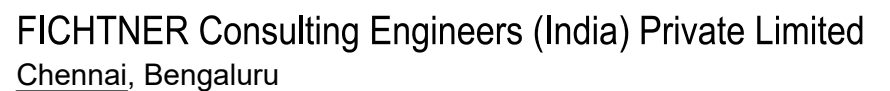
WEST AFRICAN ENERGY S.A. SENEAL NATIONAL ELECTRICITY AGENCY

EPC CONTRACTOR



CALIK ENERJİ SAN. VE TİC. A.Ş.

ENGINEERING SUBCONTRACTOR

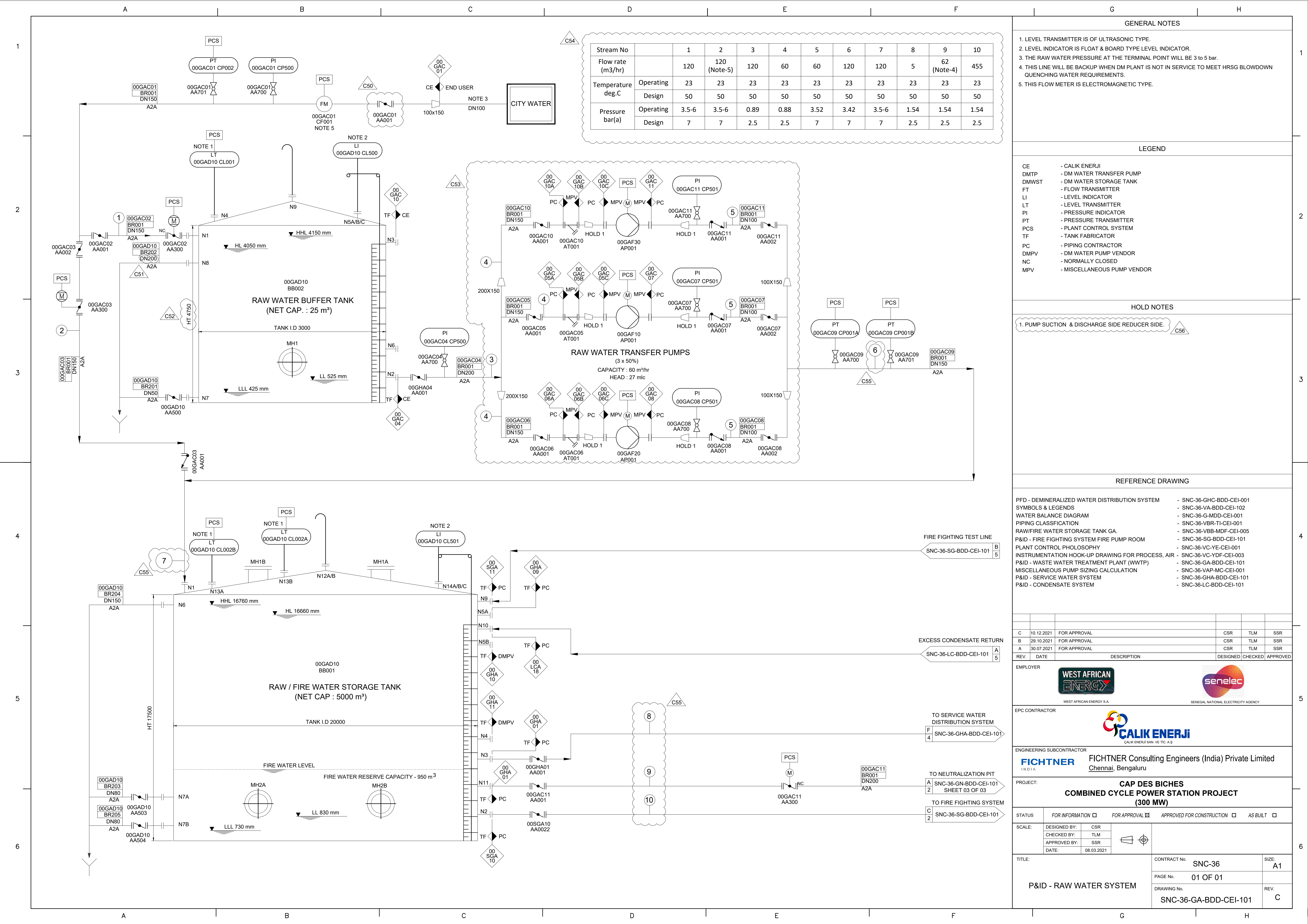


FICHTNER Consulting Engineers (India) Private Limited
Chennai, Bengaluru

PROJECT: **CAP DES BICHES
COMBINED CYCLE POWER STATION PROJECT
(300 MW)**

STATUS	FOR INFORMATION ☐	FOR APPROVAL ☐	APPROVED FOR CONSTRUCTION ☐	AS BUILT ☐
SCALE:	DESIGNED BY: RS	CHECKED BY: TLM	APPROVED BY: SSR	DATE: 25.05.2021

TITLE: P&ID - SERVICE WATER SYSTEM	CONTRACT No. SNC-36	SIZE:
	PAGE No. 01 OF 01	A1
	DRAWING No. SNC-36-GHA-BDD-CEI-101	REV. D



Stream No		1	2	3	4	5	6	7	8	9	10
Flow rate (m3/hr)		120	120 (Note-5)	120	60	60	120	120	5	62 (Note-4)	455
Temperature deg.C	Operating	23	23	23	23	23	23	23	23	23	23
	Design	50	50	50	50	50	50	50	50	50	50
Pressure bar(a)	Operating	3.5-6	3.5-6	0.89	0.88	3.52	3.42	3.5-6	1.54	1.54	1.54
	Design	7	7	2.5	2.5	7	7	7	2.5	2.5	2.5

- GENERAL NOTES
1. LEVEL TRANSMITTER IS OF ULTRASONIC TYPE.
 2. LEVEL INDICATOR IS FLOAT & BOARD TYPE LEVEL INDICATOR.
 3. THE RAW WATER PRESSURE AT THE TERMINAL POINT WILL BE 3 TO 5 bar.
 4. THIS LINE WILL BE BACKUP WHEN DM PLANT IS NOT IN SERVICE TO MEET HRSG BLOWDOWN QUENCHING WATER REQUIREMENTS.
 5. THIS FLOW METER IS ELECTROMAGNETIC TYPE.

- LEGEND
- | | |
|-------|-----------------------------|
| CE | - CALIK ENERJI |
| DMTP | - DM WATER TRANSFER PUMP |
| DMWST | - DM WATER STORAGE TANK |
| FT | - FLOW TRANSMITTER |
| LI | - LEVEL INDICATOR |
| LT | - LEVEL TRANSMITTER |
| PI | - PRESSURE INDICATOR |
| PT | - PRESSURE TRANSMITTER |
| PCS | - PLANT CONTROL SYSTEM |
| TF | - TANK FABRICATOR |
| PC | - PIPING CONTRACTOR |
| DMPV | - DM WATER PUMP VENDOR |
| NC | - NORMALLY CLOSED |
| MPV | - MISCELLANEOUS PUMP VENDOR |

- HOLD NOTES
1. PUMP SUCTION & DISCHARGE SIDE REDUCER SIDE.

- REFERENCE DRAWING
- | | |
|--|--------------------------|
| PFD - DEMINERALIZED WATER DISTRIBUTION SYSTEM | - SNC-36-GHC-BDD-CEI-001 |
| SYMBOLS & LEGENDS | - SNC-36-VA-BDD-CEI-102 |
| WATER BALANCE DIAGRAM | - SNC-36-G-MDD-CEI-001 |
| PIPING CLASSIFICATION | - SNC-36-VBR-TI-CEI-001 |
| RAW/FIRE WATER STORAGE TANK GA. | - SNC-36-VBB-MDF-CEI-005 |
| P&ID - FIRE FIGHTING SYSTEM FIRE PUMP ROOM | - SNC-36-SG-BDD-CEI-101 |
| PLANT CONTROL PHILOSOPHY | - SNC-36-VC-YE-CEI-001 |
| INSTRUMENTATION HOOK-UP DRAWING FOR PROCESS, AIR | - SNC-36-VC-YDF-CEI-003 |
| P&ID - WASTE WATER TREATMENT PLANT (WWTP) | - SNC-36-GA-BDD-CEI-101 |
| MISCELLANEOUS PUMP SIZING CALCULATION | - SNC-36-VAP-MC-CEI-001 |
| P&ID - SERVICE WATER SYSTEM | - SNC-36-GHA-BDD-CEI-101 |
| P&ID - CONDENSATE SYSTEM | - SNC-36-LC-BDD-CEI-101 |

C	10.12.2021	FOR APPROVAL		CSR	TLM	SSR
B	29.10.2021	FOR APPROVAL		CSR	TLM	SSR
A	30.07.2021	FOR APPROVAL		CSR	TLM	SSR
REV.	DATE	DESCRIPTION		DESIGNED	CHECKED	APPROVED

EMPLOYER

WEST AFRICAN ENERGY S.A. SENEAL NATIONAL ELECTRICITY AGENCY

EPC CONTRACTOR

CALIK ENERJI SAN. VE TIC. A.Ş.

ENGINEERING SUBCONTRACTOR

FICHTNER Consulting Engineers (India) Private Limited
Chennai, Bengaluru

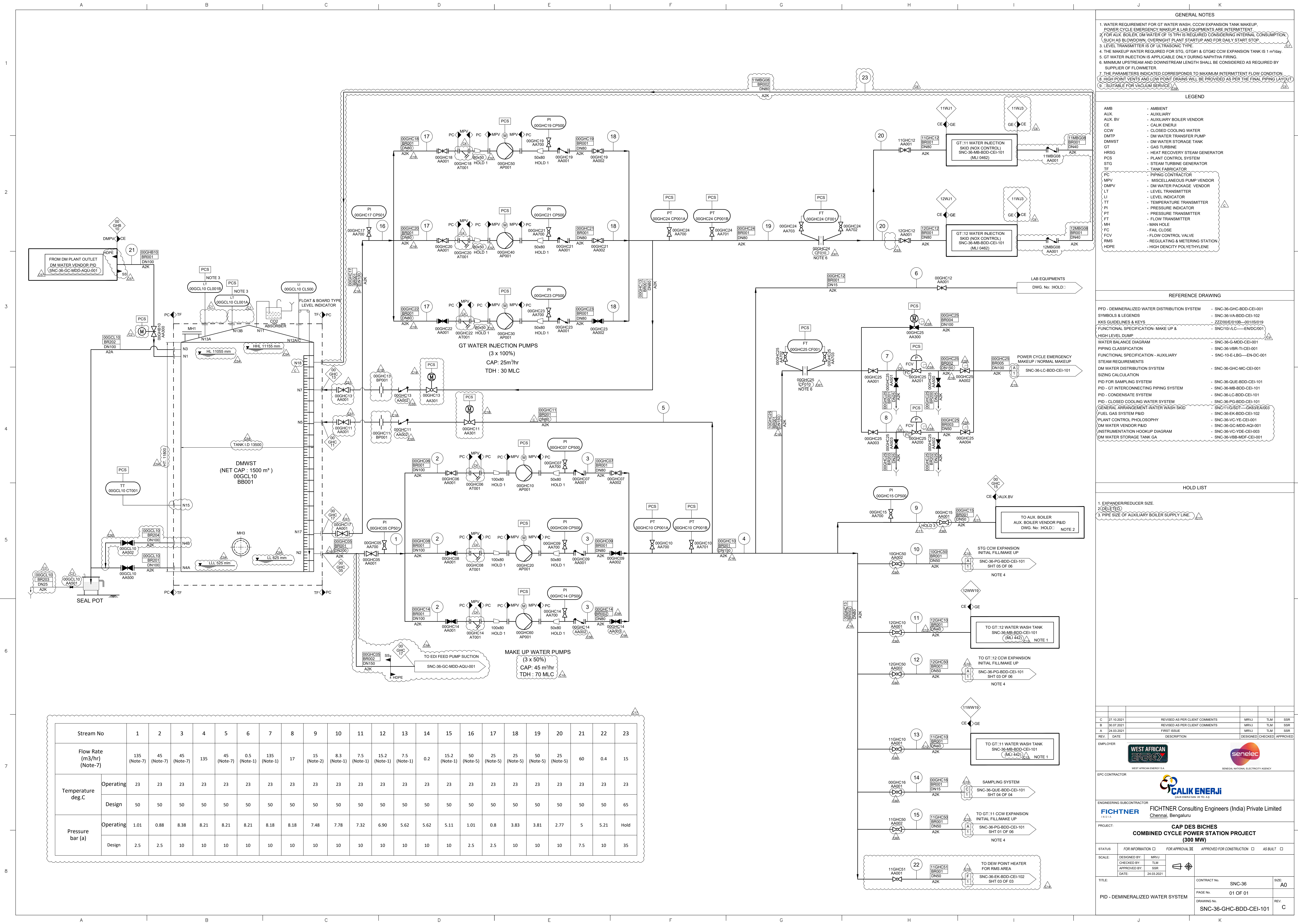
PROJECT: **CAP DES BICHES
COMBINED CYCLE POWER STATION PROJECT
(300 MW)**

STATUS: ☐ FOR INFORMATION ☒ FOR APPROVAL ☐ APPROVED FOR CONSTRUCTION ☐ AS BUILT

SCALE: DESIGNED BY: CSR
CHECKED BY: TLM
APPROVED BY: SSR
DATE: 08.03.2021

TITLE: **P&ID - RAW WATER SYSTEM**

CONTRACT No.	SNC-36	SIZE:	A1
PAGE No.	01 OF 01		
DRAWING No.	SNC-36-GA-BDD-CEI-101	REV.	C



GENERAL NOTES

1. WATER REQUIREMENT FOR GT WATER WASH, CCOW EXPANSION TANK MAKEUP, POWER CYCLE EMERGENCY MAKEUP & LAB EQUIPMENTS ARE INTERMITTENT.
2. FOR AUX. BOILER, DM WATER OF 15 TPH IS REQUIRED CONSIDERING INTERNAL CONSUMPTION, SUCH AS BLOWDOWN, OVERNIGHT PLANT STARTUP AND FOR DAILY START STOP.
3. LEVEL TRANSMITTER IS OF ULTRASONIC TYPE.
4. THE MAKEUP WATER REQUIRED FOR STG, GTG#1 & GTG#2 CCOW EXPANSION TANK IS 1 m³/day.
5. GT WATER INJECTION IS APPLICABLE ONLY DURING NAPHTHA FIRING.
6. MINIMUM UPSTREAM AND DOWNSTREAM LENGTH SHALL BE CONSIDERED AS REQUIRED BY SUPPLIER OF FLOWMETER.
7. THE PARAMETERS INDICATED CORRESPONDS TO MAXIMUM INTERMITTENT FLOW CONDITION.
8. HIGH POINT VENTS AND LOW POINT DRAINS WILL BE PROVIDED AS PER THE FINAL PIPING LAYOUT.
9. SUITABLE FOR VACUUM SERVICE.

LEGEND

AMB	- AMBIENT
AUX	- AUXILIARY
AUX. BV	- AUXILIARY BOILER VENDOR
CE	- CALIK ENERJI
CCOW	- CLOSED COOLING WATER
DMPV	- DM WATER TRANSFER PUMP
DMWST	- DM WATER STORAGE TANK
GT	- GAS TURBINE
HRSG	- HEAT RECOVERY STEAM GENERATOR
PCS	- PLANT CONTROL SYSTEM
STG	- STEAM TURBINE GENERATOR
TF	- TANK FABRICATOR
PC	- PIPING CONTRACTOR
MPV	- MISCELLANEOUS PUMP VENDOR
DMPV	- DM WATER PACKAGE VENDOR
LT	- LEVEL TRANSMITTER
LI	- LEVEL INDICATOR
TI	- TEMPERATURE TRANSMITTER
PI	- PRESSURE INDICATOR
PT	- PRESSURE TRANSMITTER
FT	- FLOW TRANSMITTER
MH	- MAN HOLE
FCV	- FAIL CLOSE
RMS	- REGULATING & METERING STATION
HDPE	- HIGH DENSITY POLYETHYLENE

REFERENCE DRAWING

PFID - DEMINERALIZED WATER DISTRIBUTION SYSTEM	- SNC-36-GHC-BDD-CEI-001
SYMBOLS & LEGENDS	- SNC-36-VA-BDD-CEI-102
JKS GUIDELINES & KEYS	- ZZZ/00/E/0108-001/S/019
FUNCTIONAL SPECIFICATION-MAKE UP &	- SNC-10/LC-EN/DC/001
HIGH LEVEL DUMP	- SNC-36-G-MDD-CEI-001
WATER BALANCE DIAGRAM	- SNC-36-LC-BDD-CEI-101
PIPING CLASSIFICATION	- SNC-36-VBR-TI-CEI-001
FUNCTIONAL SPECIFICATION - AUXILIARY	- SNC-10-E-BG-EN-DC-001
STEAM REQUIREMENTS	- SNC-36-GHC-MC-CEI-001
DM WATER DISTRIBUTION SYSTEM	- SNC-36-QUE-BDD-CEI-101
SIZING CALCULATION	- SNC-36-LC-BDD-CEI-101
PID FOR SAMPLING SYSTEM	- SNC-36-QUE-BDD-CEI-101
PID - GT INTERCONNECTING PIPING SYSTEM	- SNC-36-LC-BDD-CEI-101
PID - CONDENSATE SYSTEM	- SNC-36-PG-BDD-CEI-101
PID - CLOSED COOLING WATER SYSTEM	- SNC-36-VE-BDD-CEI-101
GENERAL ARRANGEMENT-WATER WASH SKID	- SNC-36-VE-BDD-CEI-101
FUEL GAS SYSTEM P&ID	- SNC-36-VE-BDD-CEI-101
PLANT CONTROL PHOTOSOPHY	- SNC-36-VE-BDD-CEI-101
DM WATER VENDOR P&ID	- SNC-36-VE-BDD-CEI-101
INSTRUMENTATION HOOKUP DIAGRAM	- SNC-36-VE-BDD-CEI-101
DM WATER STORAGE TANK GA	- SNC-36-VBR-TI-CEI-001

HOLD LIST

1. EXPANDER/REDUCER SIZE.
2. DELETED
3. PIPE SIZE OF AUXILIARY BOILER SUPPLY LINE.

C	27.10.2021	REVISED AS PER CLIENT COMMENTS	MRVJ	TLM	SSR
B	30.07.2021	REVISED AS PER CLIENT COMMENTS	MRVJ	TLM	SSR
A	24.03.2021	FIRST ISSUE	MRVJ	TLM	SSR
REV.	DATE	DESCRIPTION	DESIGNED	CHECKED	APPROVED

EMPLOYER: **WEST AFRICAN ENERGY** **senelec**

EPC CONTRACTOR: **CALIK ENERJI**

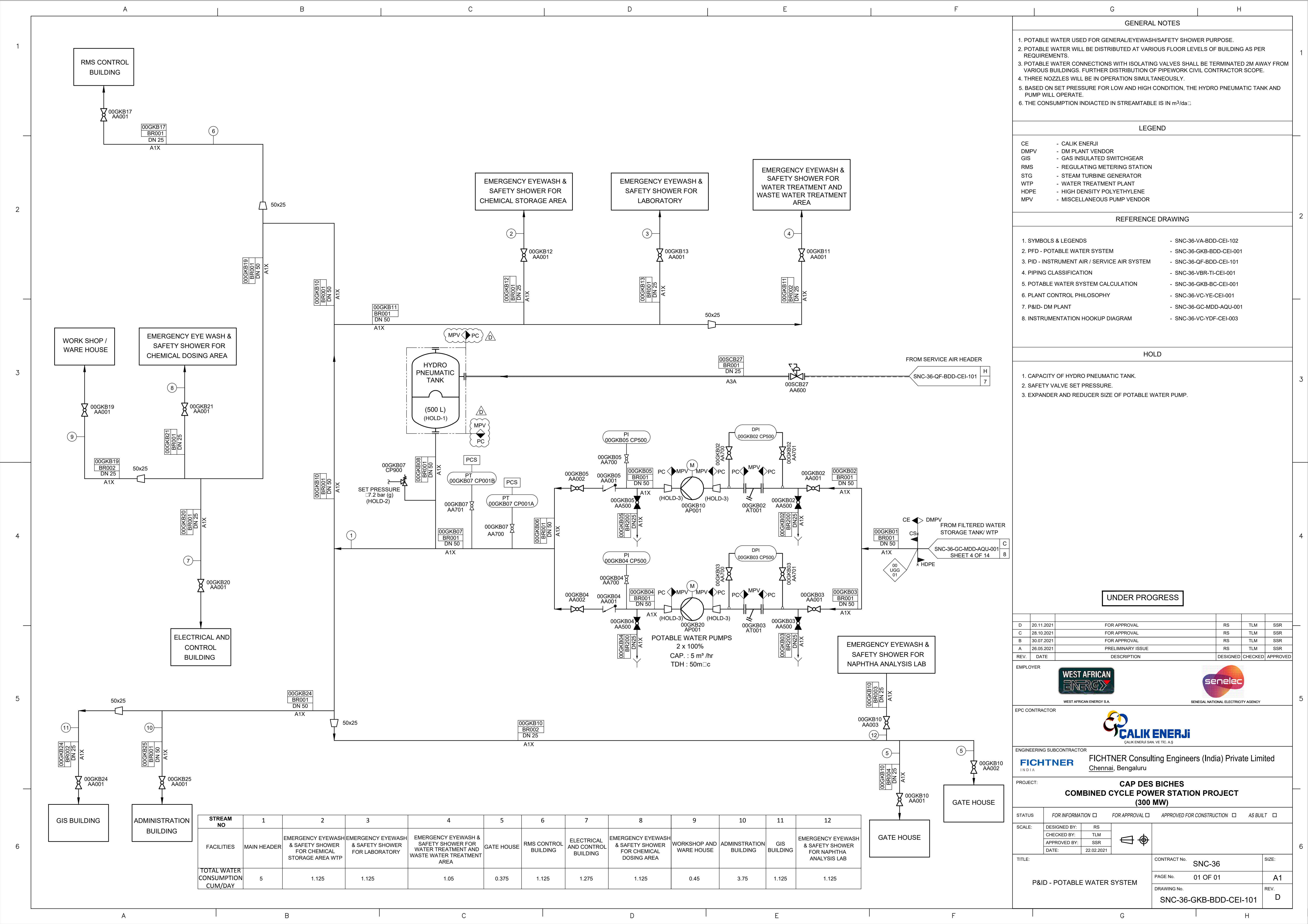
ENGINEERING SUBCONTRACTOR: **FICHTNER** **FICHTNER Consulting Engineers (India) Private Limited**
Chennai, Bengaluru

PROJECT: **CAP DES BICHES COMBINED CYCLE POWER STATION PROJECT (300 MW)**

STATUS	FOR INFORMATION ☐	FOR APPROVAL ☐	APPROVED FOR CONSTRUCTION ☐	AS BUILT ☐
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SCALE:	DESIGNED BY: MRVJ	CHECKED BY: TLM	APPROVED BY: SSR
	DATE: 24.03.2021		

TITLE: PID - DEMINERALIZED WATER SYSTEM	CONTRACT No: SNC-36	SIZE: A0
	PAGE No: 01 OF 01	
	DRAWING No: SNC-36-GHC-BDD-CEI-101	REV: C



GENERAL NOTES

- 1. POTABLE WATER USED FOR GENERAL/EYEWASH/SAFETY SHOWER PURPOSE.
- 2. POTABLE WATER WILL BE DISTRIBUTED AT VARIOUS FLOOR LEVELS OF BUILDING AS PER REQUIREMENTS.
- 3. POTABLE WATER CONNECTIONS WITH ISOLATING VALVES SHALL BE TERMINATED 2M AWAY FROM VARIOUS BUILDINGS. FURTHER DISTRIBUTION OF PIPEWORK CIVIL CONTRACTOR SCOPE.
- 4. THREE NOZZLES WILL BE IN OPERATION SIMULTANEOUSLY.
- 5. BASED ON SET PRESSURE FOR LOW AND HIGH CONDITION, THE HYDRO PNEUMATIC TANK AND PUMP WILL OPERATE.
- 6. THE CONSUMPTION INDICATED IN STREAMTABLE IS IN m³/da.

LEGEND

- CE - CALIK ENERJII
- DMPV - DM PLANT VENDOR
- GIS - GAS INSULATED SWITCHGEAR
- RMS - REGULATING METERING STATION
- STG - STEAM TURBINE GENERATOR
- WTP - WATER TREATMENT PLANT
- HDPE - HIGH DENSITY POLYETHYLENE
- MPV - MISCELLANEOUS PUMP VENDOR

REFERENCE DRAWING

- 1. SYMBOLS & LEGENDS - SNC-36-VA-BDD-CEI-102
- 2. PFD - POTABLE WATER SYSTEM - SNC-36-GKB-BDD-CEI-001
- 3. PID - INSTRUMENT AIR / SERVICE AIR SYSTEM - SNC-36-QF-BDD-CEI-101
- 4. PIPING CLASSIFICATION - SNC-36-VBR-TI-CEI-001
- 5. POTABLE WATER SYSTEM CALCULATION - SNC-36-GKB-BC-CEI-001
- 6. PLANT CONTROL PHILOSOPHY - SNC-36-VC-YE-CEI-001
- 7. P&ID- DM PLANT - SNC-36-GC-MDD-AQU-001
- 8. INSTRUMENTATION HOOKUP DIAGRAM - SNC-36-VC-YDF-CEI-003

HOLD

- 1. CAPACITY OF HYDRO PNEUMATIC TANK.
- 2. SAFETY VALVE SET PRESSURE.
- 3. EXPANDER AND REDUCER SIZE OF POTABLE WATER PUMP.

UNDER PROGRESS

REV.	DATE	DESCRIPTION	DESIGNED	CHECKED	APPROVED
D	20.11.2021	FOR APPROVAL	RS	TLM	SSR
C	28.10.2021	FOR APPROVAL	RS	TLM	SSR
B	30.07.2021	FOR APPROVAL	RS	TLM	SSR
A	28.05.2021	PRELIMINARY ISSUE	RS	TLM	SSR

EMPLOYER



WEST AFRICAN ENERGY S.A. SENEAL NATIONAL ELECTRICITY AGENCY

EPC CONTRACTOR



CALIK ENERJII SAN. VE TIC. A.Ş.

ENGINEERING SUBCONTRACTOR



FICHTNER Consulting Engineers (India) Private Limited
Chennai, Bengaluru

PROJECT:

**CAP DES BICHES
COMBINED CYCLE POWER STATION PROJECT
(300 MW)**

STATUS

FOR INFORMATION ☐ FOR APPROVAL ☐ APPROVED FOR CONSTRUCTION ☐ AS BUILT ☐

SCALE:

DESIGNED BY: RS
CHECKED BY: TLM
APPROVED BY: SSR
DATE: 22.02.2021

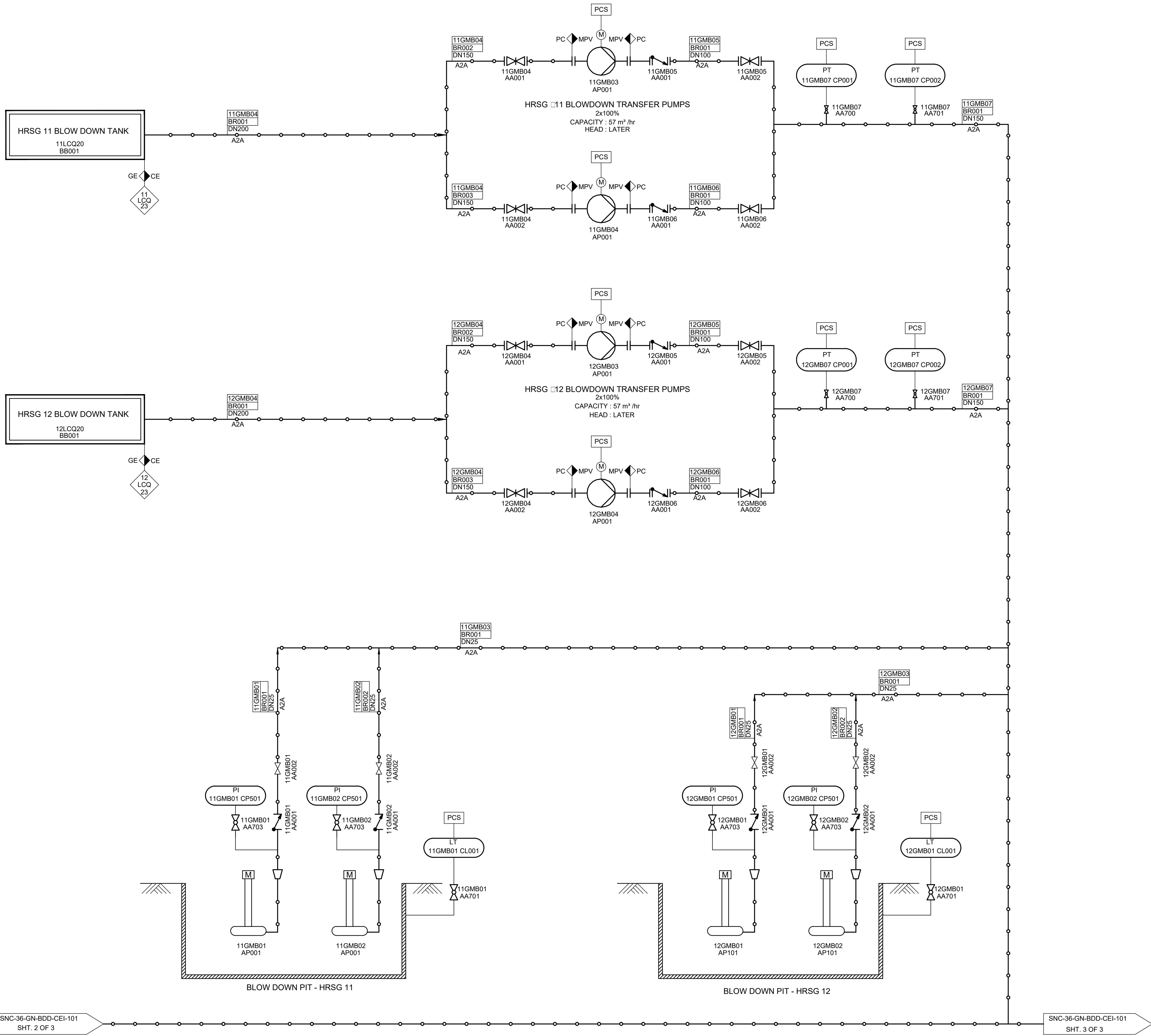


TITLE:

P&ID - POTABLE WATER SYSTEM

CONTRACT No. SNC-36
PAGE No. 01 OF 01
DRAWING No. SNC-36-GKB-BDD-CEI-101

SIZE: A1
REV: D



SNC-36-GN-BDD-CEI-101
SHT. 2 OF 3

SNC-36-GN-BDD-CEI-101
SHT. 3 OF 3

GENERAL NOTES

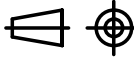
- HIGH POINT VENTS AND LOW POINT DRAINS SHALL BE PROVIDED AT SUITABLE LOCATIONS AS PER THE FINAL PIPING LAYOUT.
- ALL PRESSURE TAPPING SHALL BE OF 20NB.
- ALL PIPE LINE DRAIN & VENT SHALL BE OF 25NB SIZE WITH SINGLE ISOLATION VALVE.

LEGEND

- HRSG - HEAT RECOVERY STEAM GENERATOR
LI - LEVEL INDICATOR
LT - LEVEL TRANSMITTER
PI - PRESSURE INDICATOR
PT - PRESSURE TRANSMITTER
PCS - PLANT CONTROL SYSTEM
STG - STEAM TURBINE GENERATOR
ACC - AIR COOLED CONDENSER

REFERENCE DRAWING

- | | |
|-------------------------------------|------------------------------|
| 1. SYMBOLS & LEGENDS | - SNC-36-VA-BDD-CEI-102 |
| 2. PIPING MATERIAL SPECIFICATION | - SNC-36-VBR-TI-CEI-001 |
| 3. KKS GUIDELINES AND KEYS | - ZZZ/00/E/010b---001/IS/019 |
| 4. SNC-36-VDD-MDF-CEI-001-PLOT PLAN | - SNC-36-VA-BDD-CEI-102 |

B	08.11.2021	UNDER PROGRESS		CSR	TLM	SSR
A	30.07.2021	FOR APPROVAL		CSR	TLM	SSR
REV.	DATE	DESCRIPTION		DESIGNED	CHECKED	APPROVED
EMPLOYER						
						
WEST AFRICAN ENERGY S.A.			SENEGAL NATIONAL ELECTRICITY AGENCY			
EPC CONTRACTOR						
						
CALIK ENERJİ SAN. VE TİC. A.Ş.						
ENGINEERING SUBCONTRACTOR						
		FICHTNER Consulting Engineers (India) Private Limited Chennai, Bengaluru				
PROJECT:						
CAP DES BICHES COMBINED CYCLE POWER STATION PROJECT (300 MW)						
STATUS		FOR INFORMATION ☐ FOR APPROVAL ☒ APPROVED FOR CONSTRUCTION ☐ AS BUILT ☐				
SCALE:		DESIGNED BY: CSR				
		CHECKED BY: TLM				
		APPROVED BY: SSR				
		DATE: 08.03.2021				
TITLE:			CONTRACT No. SNC-36			SIZE: A1
P&ID - WASTE WATER TREATMENT PLANT (WWTP)			PAGE No. 01 OF 03			REV. B
			DRAWING No. SNC-36-GN-BDD-CEI-101			